

Nikos S. Soukos, DDS, PhD, MSc
1611 Lewis O Gray Dr, Saugus, MA 01906
nsoukos7@gmail.com 617-913-9763 (cell)

PROFILE

Distinguished academic teacher and mentor at Northeastern and Harvard University. Translational Medical Scientist and Entrepreneur. Pioneering Leader in the field of Photomedicine and Oral Medicine. Founding Director of the Applied Molecular Photomedicine Laboratory (the first photomedicine lab in the field of oral research worldwide) at the Harvard-affiliated Forsyth Institute. Scientific Founder and Chief Scientific Officer of PhotOral, Inc. Founder, President, and Chief Executive Officer of Photomedics Inc. Superior interpersonal skills, capable of resolving complex issues and motivating team to peak performance.

RECENT WORK EXPERIENCE

NORTHEASTERN UNIVERSITY Boston, MA
Teaching Professor (Departments of Physics & Biology, College of Science) 2015-present

- Designed and teaches the following undergraduate courses: “Medical Physics” (2011-present), “Modern Science: A journey into Matter, Life and Mind” (2015-present), “Medical Microbiology” (2015-present), and “Translational Medicine” (2018-present).

PHOTOMEDICS INC. Boston, MA
Founder, President, CEO and Chairman of the Board 2017-2023

- Co-founded a life science-based startup that developed the first intraoral light device with antimicrobial action.
- Photomedics licensed patented technology from The Forsyth Institute for translating scientific discoveries into-light based products and services in the oral health arena.
- Instrumental in the seed and series A round funding effort.
- Led a translational team of scientists, clinicians, and engineers for the completion of the product’s drawings for manufacturing and the construction of a limited number of units that can validate the design and the science behind the invention.
- Initiated a project to seek FDA approval of the product, including two meetings with FDA for obtaining approval of a proposed clinical trial protocol to test the product.
- Enriched intellectual property and expanded it into the animal market.

PHOTORAL, INC. Lexington, MA
Scientific Founder and Chief Science Officer 2011-2017

- Co-founded a life science-based startup as the commercial arm of his Applied Molecular Photomedicine Laboratory (AMPL) at The Forsyth Institute.
- Instrumental in the seed round funding effort and the key speaker in major business events.
- Designed the clinical study that led to execution of a license agreement between PhotOral and Sunstar in 2013 - a leading global company in the Mouth & Body Care field – for the development of an intraoral light device with antibacterial action.
- Winner of Mass Innovation Nights (2012).
- Finalist in Mass Challenge Accelerator (2012).

THE FORSYTH INSTITUTE (Harvard affiliated Institution) Cambridge, MA
Founding Director of the Applied Molecular Photomedicine Laboratory (AMPL) 2002-2015

- Envisioned and developed a new direction in Oral Research by establishing a unique research program in the field of Photomedicine with the foundation of AMPL (funded by NIH in 2002). The vision of AMPL is the development of novel light-guided and light-activated multifunctional systems for prevention, diagnosis, treatment and control of infectious and premalignant diseases.
- Recipient of over \$5.5 million in 20 grants from NIH and industry.
- Led and managed multi-task teams of scientists, clinicians and engineers.
- Designed and conducted original and multidisciplinary medical and scientific research leading to novel and minimally invasive technologies.

- Demonstrated the efficacy of photodynamic therapy (synergism of a photosensitizing compound and visible light) as an adjunctive technique to standard chemo-mechanical endodontic disinfection (funded by NIH in 2006).
- Established the Nano-Photo-Medicine Initiative, whose goal was to bring Photomedicine and Nanotechnology together (funded by NIH in 2009).
- Director of the “Medical Physics” course (2011-present) and of the “Introduction to Science” course (2015-) for undergraduate students at Northeastern University.
- Recipient of the Excellence in Teaching Award, Harvard Medical School (2003).

WELLMAN LABORATORIES OF PHOTOMEDICINE, DEPARTMENT OF DERMATOLOGY, MASSACHUSETTS GENERAL HOSPITAL, HARVARD MEDICAL SCHOOL

Boston, MA

Research Scientist & Instructor

1998-2002

- Led the investigation of drug delivery into bacterial biofilms.
- Co-Investigator in the first clinical trials in US on photodynamic therapy of oral precancer.
- As key Scientific Expert for the Office of Chief Medical Examiner, State of Massachusetts, contributed to the development of a rapid fluorescence spectroscopic method to detect dried saliva stains swabbed from human skin in forensic trace evidence.
- Fellow of the Boston Oral Health Clinical Resource Center (1998-1999).
- Recipient of the Culpeper Biomedical Pilot Initiative Award, Rockefeller Brothers Fund (2001-2002).

Post-doctoral Research Fellow

1995-1998

- Received multidisciplinary training in photobiology, photochemistry, photophysics, cell biology and microbiology.
- Led the first study ever on the development of immunophotodiagnosis into an accurate screening and diagnostic tool for early detection of oral precancer.
- Investigated the development of a novel photochemotherapeutic system for the treatment of periodontal disease.
- Fellow of the MGH Laser Center (1996-1998).

EARLIER EXPERIENCE

EASTMAN DENTAL INSTITUTE, DEPARTMENT OF ORAL PATHOLOGY, UNIVERSITY OF LONDON, UK

London, MA

Post-doctoral Research Fellow

1992-1995

- Investigated photodynamic therapy on oral infection and cancer.
- Managed the Cell Culture Laboratory and trained students in cell culture techniques.

GENERAL AIRFORCE HOSPITAL, DEPT. OF ORAL MEDICINE

Athens, Greece

Clinical Fellow

1989-1991

- Worked as a full-time member of the clinical staff.

UNIVERSITY OF MUNICH, CLINIC OF ORAL AND MAXILLOFACIAL SURGERY & INSTITUTE OF RADIOBIOLOGY

Munich, Germany

Doctoral Research Fellow

1985-1989

- Conducted basic research on cell kinetics of normal oral mucosa and oral precancer.
- Received clinical training in oral medicine.

APPOINTMENTS

2023 -

Teaching Professor, Departments of Physics and Biology, College of Science, Northeastern University, Boston.

2019 - 2023

Associate Teaching Professor, Departments of Physics and Biology, College of Science, Northeastern University, Boston.

2019 - 2023	Assistant Teaching Professor, Departments of Physics and Biology, College of Science, Northeastern University, Boston.
2017 - 2023	Founder, President, CEO and Chairman of the Board, Photomedics Inc., Boston, Massachusetts.
2011 - 2017	Chief Science Officer of PhotOral, Inc. Lexington, Massachusetts.
2014 - 2015	Senior Research Investigator (equivalent to Professor), Forsyth Institute.
2008 - 2014	Associate Member of the Staff (equivalent to Associate Professor), Forsyth Institute.
2002 - 2008	Assistant Member of the Staff (equivalent to Assistant Professor), Forsyth Institute.

EDUCATION

1992	M.Sc. in Experimental Oral Pathology, Eastman Dental Institute, University of London, United Kingdom
1989	Ph.D. (Biology of Oral Precancer), University of Munich, Germany
1984	D.D.S. Dental School of Athens, Greece